

Navigators IDR Simulator - Button Documentation

This document provides a comprehensive list of all interactive buttons within the Navigators IDR Simulator interface, detailing their identifiers, their function, and the underlying JavaScript components they are connected to.

Navigation & Sidebar Buttons

1. Workspace Tabs

- **Identifier:** Class `.nav-link`
- **Label:** "Navigation", "Saved playback", "Experiments", "Device timings"
- **Function:** Switches between the different application views/workspaces.
- **Connection:** Connected in `workspace.js`. Clicking triggers the `setView(viewId)` function which toggles the `hidden` attribute on the corresponding `<section>` elements.

2. Demo field guide

- **Identifier:** ID `#btnGuide`
- **Label:** "Demo field guide ↗"
- **Function:** Opens the instructions modal dialog.
- **Connection:** Connected in `workspace.js`. Triggers `dialog.showModal()`.

3. Close Guide

- **Identifier:** ID `#btnCloseGuide`
- **Label:** "x"
- **Function:** Closes the instructions modal dialog.

- **Connection:** Connected in `workspace.js`. Triggers `dialog.close()`.



Live Navigation & Map Panel

4. Start Offline Engine / Stop Engine

- **Identifier:** ID `#btnStartLive`
- **Label:** "Start Offline Engine" (changes to "Stop Engine" when running)
- **Function:** Toggles the state of the local navigation engine. When started, it requests sensor permissions, loads the AI models, begins capturing data, and updates the UI status indicators.
- **Connection:** Connected in `app.js`. Interacts with `window.offlineEngine`.

5. Simulate GNSS Outage / Restore GPS

- **Identifier:** ID `#btnGpsOutage`
- **Label:** "Simulate GNSS outage" (changes to "Restore GPS")
- **Function:** Toggles the `engine.gpsOutage` boolean flag. When simulated outage is active, the system ignores live GPS coordinates and falls back to local Dead Reckoning to estimate position.
- **Connection:** Connected in `app.js`. Interacts with `window.offlineEngine`.

6. Center local map

- **Identifier:** ID `#btnFitArea`
- **Label:** Target icon inside map panel
- **Function:** Centers the Leaflet map view either to the saved offline bounding box or to the current GPS location.
- **Connection:** Connected in `workspace.js`. Triggers `fitLocalArea()`.

7. Save area offline

- **Identifier:** ID `#btnSaveArea`

- **Label:** "Save area offline"
- **Function:** Grabs the current map bounds, identifies the required map tiles, and downloads them into the browser's IndexedDB. Allows the map to render when internet connectivity is lost.
- **Connection:** Connected in `workspace.js`. Interacts with the `LocalMap` instance and IndexedDB APIs.

8. Follow position

- **Identifier:** ID `#btnFollow`
- **Label:** "Follow position"
- **Function:** Toggles whether the map view automatically pans to keep the user's current position marker centered.
- **Connection:** Connected in `workspace.js`. Toggles the `state.followPosition` flag.

Playback Controls

9. Play Trajectory

- **Identifier:** ID `#btnPlay`
- **Label:** Play icon
- **Function:** Starts the playback loop of a loaded trajectory JSON file.
- **Connection:** Connected in `app.js`. Triggers the `play()` function.

10. Pause Trajectory

- **Identifier:** ID `#btnPause`
- **Label:** Pause icon
- **Function:** Pauses the currently playing trajectory.
- **Connection:** Connected in `app.js`. Triggers the `pause()` function.

11. Reset Trajectory

- **Identifier:** ID #btnReset
- **Label:** "↺" (Reset icon)
- **Function:** Resets the trajectory playback index to 0 and updates the timeline slider to the beginning.
- **Connection:** Connected in app.js. Triggers the reset() function.

PC Sync & App Settings

12. Pair PC

- **Identifier:** ID #btnPairPC
- **Label:** "Pair PC"
- **Function:** Reads the pairing code from the input and attempts to establish a connection with the local Python server for syncing recorded data.
- **Connection:** Connected in workspace.js. Interacts with the RecordingSync class.

13. Retry Sync

- **Identifier:** ID #btnSyncPC
- **Label:** "Retry sync"
- **Function:** Attempts to manually flush any locally buffered sensor data to the connected PC if automatic background sync failed.
- **Connection:** Connected in workspace.js. Calls window.recordingSync.flush().

14. Update App

- **Identifier:** ID #btnUpdateApp
- **Label:** "Update app"
- **Function:** Interacts with the browser's Service Worker to fetch the latest cached files and reloads the window to apply the update.
- **Connection:** Connected in workspace.js. Accesses navigator.serviceWorker.

15. Start ML Training

- **Identifier:** ID #btnStartTraining
- **Label:** "Start ML Training"
- **Function:** Sends a request to the paired backend to initiate model training using the synced recording data.
- **Connection:** Connected in app.js and mac_dashboard.js. Triggers the startTraining() function.



Performance Profiling

16. Reset Timings

- **Identifier:** ID #btnResetTimings
- **Label:** "Reset timings"
- **Function:** Clears the currently recorded performance metrics/timings for device profiling.
- **Connection:** Connected in app.js. Interacts with window.deviceProfiler.

17. Export Device Timings

- **Identifier:** ID #btnExportTimings
- **Label:** "Export device timings"
- **Function:** Serializes the device profiler stats into a JSON Blob and triggers a local file download.
- **Connection:** Connected in app.js. Calls window.deviceProfiler.getStats().



Startup Privacy Modal

18. Accept & Continue

- **Identifier:** ID #btnStart

- **Label:** "Accept & Continue"
- **Function:** Confirms acceptance of terms, closes the modal, sets a consent flag in `localStorage`, and initializes the workspace.
- **Connection:** Connected in `workspace.js`. Interacts with DOM class lists and `localStorage`.

19. Reject All

- **Identifier:** ID `#btnSkip`
- **Label:** "Reject All"
- **Function:** Plays a shake animation and changes text to "Please accept T&C" to enforce consent.
- **Connection:** Connected in `workspace.js`. Modifies DOM styles and classes.